

QUEST INNOVATIVE SOLUTIONS

PYTHON FULL STACK DEVELOPMENT

PYTHON CORE EXAM (SESSION 1)

Max. Marks: 50

Duration: 2 Hours

Part A

(2 marks)

1. Differentiate between the 'is' operator and the '==' operator in Python.
2. Differentiate between the 'for' loop and the 'while' loop in Python.
3. What are operators in Python?
4. What is the 'replace()' method in Python strings? Explain.
5. Differentiate between mutable and immutable objects in Python. Give two examples.
6. Explain the differences between a list and a tuple in Python. Provide at least three points for each, highlighting their characteristics and use cases.
7. How can you create a tuple containing only a single number 10 in Python?
8. How can you create an empty dictionary and an empty set in Python?
9. Define a local variable and global variable in Python. State one key difference between a local variable and a global variable.
10. Explain the main difference between pass by value and pass by reference.
11. State one key difference between *args and **kwargs.
12. What is recursion in Python?
13. What is polymorphism in Python?
14. What is encapsulation in Python?
15. What is a constructor and a destructor in Python? Explain the difference between them.
16. What is a lambda function in Python? State the syntax of a lambda function.
17. What are file modes in Python?
18. How can you delete a file in Python?
19. What is class and object in Python?
20. State one key difference between a library, a module, and a package in Python.

Part B

(5 marks)

21. Explain inheritance in Python and describe the different types of inheritance with examples.
22. Explain exception handling in Python in detail. Include the following points in your answer:
- What is an exception?
 - The purpose of exception handling.
 - How to handle multiple exceptions.
 - Give an example scenario where exception handling is useful.